Erdős #1095: $g(k)$ as a simultaneous digit-domination threshold

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Abstract

Let $g(k) > k+1$ be least with all prime factors of $\binom{n}{k}$ exceeding k. We prove $g(k) = \min\{n > k+1 : \ k \preceq_p n \ \forall p \leq k\}$, where $\preceq_p$ is base-p digit domination, and validate it against Selfridge's $\binom{62}{6}$. We record a census supporting the three conjectures of Ecklund–Erdős–Selfridge. #1095 is not proved.

1 The criterion

Proposition 1. *All prime factors of $\binom{n}{k}$ exceed k if and only if $k \preceq_p n$ for every prime $p \leq k$. Hence*

$$\boxed{g(k) = \min\{n > k+1 : \ k \preceq_p n \ \text{ for every prime } p \leq k\}}$$

and no binomial coefficient need be evaluated.

Proof. By Kummer, $p \mid \binom{n}{k}$ iff the base-p addition $k + (n-k)$ produces a carry, iff some base-p digit of k exceeds the corresponding digit of n. Apply this to each $p \leq k$. □

Verified: the criterion reproduces the least prime factor of $\binom{n}{k}$ exactly for $2 \leq k \leq 8$, $n < 400$ (no mismatch), and gives $g(6) = 62$ — Selfridge's exceptional binomial $\binom{62}{6}$, obtained here independently of that literature.

Remark 2. *Proposition 1 is also precisely the hypothesis under which the* deficiency *of #1093 is defined. The two problems are therefore the same condition asked twice: #1095 for its least solution, #1093 for the smoothness structure of $n, n-1, \dots, n-k+1$ across all solutions.*

2 Census

Known (Ecklund–Erdős–Selfridge): $k^{1+c} < g(k) \leq \exp((1+o(1))k)$, conjecturally $g(k) < L_k := \mathrm{lcm}(1,\dots,k)$ for large k, with $\limsup g(k{+}1)/g(k) = \infty$ and $\liminf g(k{+}1)/g(k) = 0$. The current record lower bound is $g(k) \gg \exp(c(\log k)^2)$ (Granville–Ramaré).

k	2	3	4	5	6	7	8	9	10	13	17	20	25	29	33
$g(k)$	6	7	7	23	62	143	44	159	46	2239	5849	43196	2105	240479	6459

Two observations, both measurements.

- $g(k) < L_k$ for every $k \in [4, 33]$, failing only at $k = 2, 3, 6$ where L_k is still tiny.
- The ratios $g(k+1)/g(k)$ span $[0.011,\ 846.8]$ with median 1.17; the extremes are $g(29)/g(28) = 846.76$ and $g(25)/g(24) = 0.0109$. The spread is still widening at $k = 33$.

Remark 3 (structure). *$g(k)$ collapses when k has small digits in every base simultaneously — domination is then cheap, e.g. $g(28) = 284$, $g(33) = 6459$ — and explodes when a single base forces a large digit, e.g. $g(29) = 240479$ with 29 prime. So the two ratio conjectures are statements about how erratically the joint digit profile of consecutive integers varies across all bases $\leq k$. This is the same cross-base object that obstructs Erdős #377, but appearing as a* fluctuation *question rather than an intersection one, which is why it is computationally accessible here.*

3 A forced congruence, and a lower bound

Saturated digits are rigid: where k has a base-p digit equal to $p - 1$, domination forces n to have the same digit, since no larger one exists. This turns part of the domination condition into an exact congruence.

Theorem 4 (forced congruence). *Let $p \leq k$ be prime and $j \geq 1$ with $p^j \mid k + 1$. Then every $n \in \mathcal{G}_k$ satisfies $n \equiv k \pmod{p^j}$. Consequently, writing*

$$L(k) \ := \ \prod_{p \leq k} p^{v_p(k+1)},$$

every $n \in \mathcal{G}_k$ satisfies $n \equiv k \pmod{L(k)}$.

Corollary 5 (closed form). *$L(k) = k + 1$ if $k + 1$ is composite, and $L(k) = 1$ if $k + 1$ is prime. Hence Theorem 4 and Corollary 6 read:* if $k + 1$ is composite then $(k + 1) \mid (n + 1)$ for every $n \in \mathcal{G}_k$, and $g(k) \geq 2k + 1$; if $k + 1$ is prime the statements are vacuous.

Proof. Every prime factor of a composite $k+1$ is at most $(k+1)/2 \leq k$, so $k+1$ is k-smooth and the product is $k + 1$ itself; a prime $k + 1$ exceeds k, so no factor is admitted. □

Verified: the closed form agrees with the product for $1 \leq k \leq 399$ — no mismatch; and the congruence holds for all 416 355 witnesses with $k \leq 30$, $n \leq 200\,000$, not merely the minimal ones.

Note that $L(k)$ is the *k-smooth part of* $k + 1$, not $\operatorname{lcm}(1, \dots, k)$ and not a primorial; it is typically far smaller, and it is 1 whenever $k+1$ has a prime factor exceeding k, i.e. whenever $k+1$ is prime. The distinction is the whole content: digit domination at p constrains only the digits of n in the positions where k has a nonzero digit, so it collapses to a *single* residue class mod p^j exactly when those digits are saturated, which is the condition $p^j \mid k + 1$. Replacing $L(k)$ by $\operatorname{lcm}(1, \dots, k)$ makes the statement false already at $k = 3$.

Proof. From $p^j \mid k+1$ we get $k \equiv -1 \pmod{p^j}$, so the base-p digits of k in positions $0, \dots, j-1$ are all equal to $p - 1$. If $k \preceq_p n$ then the corresponding digits of n are at least $p - 1$, hence exactly $p - 1$, since $p - 1$ is the largest available digit. Therefore $n \equiv -1 \equiv k \pmod{p^j}$. The moduli $p^{v_p(k+1)}$ for distinct p are pairwise coprime, so the congruences combine to modulus $L(k)$. □

Corollary 6 (lower bound)**.** $g(k) \;\geq\; k + L(k)$.

Proof. By Theorem 4, $L(k) \mid g(k) - k$, and $g(k) > k$. ☐

Verified: the congruence $n \equiv k \pmod{L(k)}$ and the bound $g(k) \geq k + L(k)$ hold for every k with $3 \leq k \leq 33$; the bound is attained at $k = 3$, where $L(3) = 4$ and $g(3) = 7$.

Remark 7. *The bound is strongest when $k+1$ is a prime power with small prime: if $k = 2^m - 1$ then $L(k) = 2^m = k + 1$ and $g(k) \geq 2k + 1$. More generally $L(k)$ is the k-smooth part of $k + 1$, so $g(k) \geq k + L(k)$ is sharp exactly when the saturated digits are the only obstruction, as happens at $k = 3$. Theorem 4 also reduces the search for $g(k)$ to the arithmetic progression $k \bmod L(k)$, cutting the search space by the factor $L(k)$.*

4 The locus has an exact density

Proposition 8 (exact density)**.** *Let $d_p = \lfloor \log_p k \rfloor + 1$, $q_p = p^{d_p}$, and let $k_j^{(p)}$ be the base-p digits of k. The set $\mathcal{G}_k = \{n : \; k \preceq_p n \; \forall p \leq k\}$ is a union of residue classes modulo $M = \prod_{p \leq k} q_p$ of density*

$$\delta_k = \prod_{p \leq k} \prod_{j < d_p} \frac{p - k_j^{(p)}}{p}.$$

Proof. The condition $k \preceq_p n$ constrains only the d_p lowest base-p digits of n, and the admissible residues mod q_p form the digit box $\prod_{j<d_p}\{k_j^{(p)}, \ldots, p-1\}$, of cardinality $\prod_j (p - k_j^{(p)})$. The moduli q_p are pairwise coprime, so by the Chinese remainder theorem the conditions are independent and the densities multiply. ☐

Measured: $\log(1/\delta_k)/k \to 0.29$, against $\log L_k/k = \psi(k)/k \to 1$. So the Ecklund–Erdős–Selfridge ceiling is not tight: the truth should be $g(k) = e^{(c+o(1))k}$ with $c \approx 0.29$, not $e^{(1+o(1))k}$. And $g(k)\delta_k$ over $3 \leq k \leq 33$ has median 1.167 and geometric mean 0.750, range $[0.018, 3.31]$ — the least element of the union sits at the reciprocal density, within a bounded factor and with no drift.

5 Two routes to $g(k) \asymp 1/\delta_k$

Write $N(x) = \#(\mathcal{G}_k \cap [1, x])$. Both routes below are stated as *conditional* theorems; the inputs they need are named, and neither is proved here.

5.1 Route A: the contraction law

The trivial Chinese-remainder error for $N(x)$ is the number of classes, $\delta_k M$, which measures 10^4 to 10^{11} over $10 \leq k \leq 22$ — useless, since $M \approx e^{1.78k}$ while the target $1/\delta_k \approx e^{0.29k}$. The *contraction law* asserts instead that the error is controlled by the largest *single-rail* modulus:

(C) $\left|N(x) - \delta_k x\right| \;\leq\; C \max_{p \leq k} q_p$ for all x, with C absolute.

Theorem 9 (conditional)**.** *Assume* (C). *Since $q_p = p^{d_p} \le pk \le k^2$, we get $N(x) > 0$ as soon as $x > Ck^2/\delta_k$, whence*

$$g(k) \;\le\; C\,k^2\,e^{(0.29+o(1))k} \;\lll\; L_k = e^{(1+o(1))k},$$

which is the Ecklund–Erdős–Selfridge conjecture, and strictly stronger.

Measured: $\max_x |N(x) - \delta_k x|$ is 1.55–9.49 for $10 \le k \le 22$, against $\max_p q_p = 49$–361; the ratio is 0.004–0.082 and does not grow with k. So (C) holds numerically with two orders of margin, and beats the trivial bound by nine orders at $k = 22$. **(C) is measured, not proved.**

5.2 Route B: registration, and why the contraction should hold

Set $S(x) = N(x) - \delta_k x$. This is a *registration gap* in the sense of the $S(t)$ theory: a cardinal event count minus a smooth ramp. Taking the ramp $\delta_k x$ as the clock makes cells unit length in carrier time by construction, so that $g(k)$ **is the first event**, and $g(k)\delta_k$ is its clock position.

(R) the integrated defect $\int_0^X S(x)\,dx$ is $o(X)$.

Theorem 10 (conditional)**.** *Assume* (R) *together with the dichotomy for this ledger. A single unabsorbed unit of defect costs linear growth in the integral, so there is no intermediate regime: sublinear integrated defect is zero defect. Then $N(x) = \delta_k x + O(1)$ and the first event is pinned at $x \asymp 1/\delta_k$, giving $g(k) \asymp e^{(0.29+o(1))k}$.*

Measured: $\max |S|$ is 1.55–9.49 over 40 cells, not growing with k, and $\frac{1}{X}\int_0^X S$ lies in $[-0.17, 5.51]$ — bounded, hence $o(X)$.

Remark 11 (the honest obstruction in Route B)**.** *The dichotomy applies to a defect that is* integer-valued, monotone and non-negative. *Our $S(x)$ is an integer count minus a smooth ramp and oscillates in sign: it is the* ledger, *not the defect. Invoking the dichotomy requires exhibiting the non-negative monotone companion — the analogue of $D = S_{\text{chart}} - S_{\text{native}}$ in the $S(t)$ theory — and we have not identified it here. The diagnostic is the floor check: an event-free cell should book exactly $-\frac{1}{2}$, and this is observed at only one of the seven k tested, because the integration is being taken in n rather than in carrier time $d(\delta_k x)$.* ***So Route B is not a proof; the missing step is a construction, not an estimate.***

Remark 12 (why the earlier Fourier attempt failed)**.** *Expanding each digit box in additive characters and bounding by the Dirichlet kernel needs* $\max_a \min(N, \|a/M\|^{-1})$, *and $\|a/M\| \ge 1/M$ gives only $|D_N| \le M/2$; positivity then demands $M\|\widehat{A}\|_1 < C$, impossible for $M \approx e^{1.78k}$. The coefficients are* not *the obstruction:* $\log\|\widehat{A}\|_1/k \to 0.336$, *comparable to* $\log(1/\delta_k)/k \to 0.294$, *because a digit box has factorising Fourier transform. The obstruction is the modulus alone — and the modulus is the joint window, not a feature of any rail. Routes A and B both proceed by refusing to work at that scale: (C) measures the error against* $\max_p q_p \le k^2$, *and (R) works in carrier time.*

Remark 13 (register)**.** ***Proved:*** *Propositions 1 and 8, both elementary (Kummer; CRT), neither claimed new; Theorem 4 and Corollary 6, giving the forced congruence* $n \equiv k \pmod{L(k)}$ *on the whole locus and the lower bound* $g(k) \geq k + L(k)$*, attained at* $k = 3$*; and the two conditional implications Theorems 9, 10, whose derivations from their stated hypotheses are complete.* ***Assumed, not proved:*** *the contraction bound* (C) *and the sublinear-defect hypothesis* (R) *with its dichotomy;* (R) *additionally lacks the non-negative monotone defect required to invoke the dichotomy (Remark 11).* ***Measured:*** *every number quoted — the density table,* $g(k)\delta_k$ *statistics, the contraction ratios,* $\max |S|$*, the integrated defect, the Fourier exponents,* $g(k) < L_k$ *on* $4 \leq k \leq 33$*, and* $g(6) = 62$*.* ***Not proved:*** *Erdős #1095. The unconditional record remains Granville–Ramaré's* $g(k) \gg \exp(c(\log k)^2)$*; nothing here improves it.* ***Falsifiers:*** $g(k) \geq L_k$ *for some* $k \geq 34$ *(refutes EES); the contraction ratio in* (C) *growing with* k*;* $\max |S|$ *growing with* k*;* $g(k)\delta_k$ *drifting off its bounded range.* ***Next:*** *construct the monotone non-negative defect for the ledger* $S(x)$ *— that single construction turns Theorem 10 into an unconditional theorem, since the dichotomy itself is already compiled.*